Zero output of a perturbed system

Let A be a symmetric invertible matrix of size $n \times n$. Typically, we can use $y = Ax$ to represent a linear relationship between y (as the output) and x (as the input). When A is invertible, the only vector x that gives zero output is the zero input vector (verify this fact.) However, in this problem, our objective is to find a scalar $c \in \mathbf{R}$ such that the perturbed system

$$y = (A + cI)x$$

always has a *non-zero* input vector x that is mapped to the zero output.

Problem

1. For a given A, your task is to find c and a non-zero x that makes the zero output. If you have many solutions in x (for each value of c), pick the one that has $\|x\| = 5$. Also return the number of distinct values of c in variable num_c.
2. How many values of c can you find? If there are many values of c and corresponding value of x, stack all possible answers in an array. For example, if there are 3 values of c, return $c = (c_1, c_2, c_3)$ as an array, set $num_c = 3$. Also return $X = [x_1\ x_2\ x_3]$ as a matrix of 3 columns where $x_k \in \mathbf{R}^n$. If you think there are infinitely many values of c, return the parameter $\mathbf{num_c} = \mathbf{np.inf}$ but give only a finite number of c.

Implementation

Goal: Fill the given function then submit your function in grader:

```
import numpy as np

def zero_output_perturbedsys(A):
    ...
    return c, X, num_c
```

Constraints

Input:

- $A \in \mathbf{R}^{n \times n}$: symmetric and invertible with $10 \leq n \leq 500$

Output:

- $c \in \mathbf{R}^K$
- $X \in \mathbf{R}^{n \times K}$
- num_c as K or $np.inf$